# George Stepaniants

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I have broad interests in scientific machine learning, computer-assisted mathematics, dynamical systems, and optimal transport. My research develops methods for efficient and accurate model inference and prediction in complex physical systems, backed by rigorous analysis, approximation bounds, and statistical guarantees.

## Academic Employment

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| Oct 2026 – Present | **University of Cambridge (Assistant Professor)**<br>*Assistant Professor in Mathematics of Information*<br>Department of Applied Mathematics and Theoretical Physics (DAMTP) |
| Oct 2026 – Present | **New York University (Visiting Faculty)**<br>Courant Institute School of Mathematics, Computing, and Data Science |
| Sep 2024 – 2026 | **California Institute of Technology (Postdoctoral Scholar)**<br>*NSF Mathematical Sciences Postdoctoral Research Fellow (MSPRF)*<br>Department of Computing and Mathematical Sciences (CMS)<br>Postdoctoral Advisor: Andrew M. Stuart |

## Education

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| Sep 2019 – Jun 2024<br>**GPA**: 4.9/5.0 | **Massachusetts Institute of Technology (PhD)**<br>Department of Mathematics and Institute for Data, Systems, and Society (IDSS)<br>PhD Advisors: Philippe Rigollet and Jörn Dunkel<br>Graduate Thesis: *Inference from limited observations in statistical, dynamical, and functional problems* |
| Sep 2015 – Jun 2019<br>**GPA**: 3.87/4.00 | **University of Washington (BSc)**<br>Department of Mathematics and Department of Computer Science (double major)<br>Undergraduate Research Advisors: Nathan Kutz and Bing Brunton<br>Research Topic: *Inferring causal networks of dynamical systems through transient dynamics and perturbation* |

## Academic Awards

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| Isaac Newton Institute 2026 Simons Fellow | Aug 2026 |
| ILAS 2026 Early Career NSF Conference Grant Award Recipient | Apr 2026 |
| NSF Mathematical Sciences Postdoctoral Research Fellowship (MSPRF) 2402074 | Sep 2024 – 2026 |
| IMS Lawrence D. Brown Ph.D. Student Award Recipient 🖇 (3 recipients nationwide) | Sep 2024 |
| NSF Graduate Research Fellowship (GRFP) 1745302 | Jun 2019 – Jun 2024 |
| SIAM Student Travel Award | Dec 2023 |
| Calouste Gulbenkian Foundation Short Term Conference and Travel Grant | Jun 2023 |
| MIT Presidential Fellowship | Sep 2019 – Jun 2020 |
| Phi Beta Kappa Honors Society Member | Jun 2019 |
| Mary Gates Research Scholarship (merit-based) | Jun 2019 |
| University of Washington Dean's List | Sep 2015 – Jun 2019 |
| Early acceptance to University of Washington (UW Academy) | Sep 2015 |

## Publications

Please see also my Google Scholar and ORCID ID. († alphabetical order, * joint first author)

**Thesis**

[T1] 🖇 George Stepaniants. "Inference from Limited Observations in Statistical, Dynamical, and Functional Problems." *Massachusetts Institute of Technology* (2024).

### Preprints

[PP1] Matthew J. Colbrook and George Stepaniants. "An affine reverse Faber-Krahn inequality via information decay." *arXiv preprint arXiv:2611.02279* (2026).

[PP2] Matthew J. Colbrook and George Stepaniants. "Spectral isometries of unital C*-algebras." *arXiv preprint arXiv:2610.40470* (2026).

[PP3] Matthew J. Colbrook, George Stepaniants, and Alex Townsend. "A Complete Resolution of Forsythe's Conjecture for Restarted Conjugate Gradients." *arXiv preprint arXiv:2609.04659* (2026).

[PP4] Matthew J. Colbrook, Siavash Sadeghi, and George Stepaniants. "A computer-assisted counterexample to the planar Berenstein conjecture." *arXiv preprint arXiv:2608.08953* (2026).

[PP5] Matthew J. Colbrook, George Stepaniants, and Alex Townsend. "A Proof of the Forsythe Conjecture for the Two-Step Restarted Conjugate Gradient Method." *arXiv preprint arXiv:2608.02852* (2026).

[PP6] Matthew J. Colbrook and George Stepaniants. "A computer-assisted counterexample to the planar Pompeiu and Schiffer conjectures." *arXiv preprint arXiv:2608.01579* (2026).

### Journal Articles

[J1] Kaushik Bhattacharya, Lianghao Cao, † George Stepaniants, Andrew M. Stuart, and Margaret Trautner. "Learning Memory and Material Dependent Constitutive Laws." *The SMAI Journal of Computational Mathematics* 12 (2026): 219-267. † Authors listed alphabetically.

[J2] David Darrow and † George Stepaniants. "A Spectral Theory of Scalar Volterra Equations." *Communications of the American Mathematical Society* 6 (2026): 634-725. † Authors listed alphabetically.

[J3] Yanjun Han, Philippe Rigollet, and † George Stepaniants. "Covariance alignment: from maximum likelihood estimation to Gromov-Wasserstein." *SIAM Journal on Mathematics of Data Science* 7.3 (2025): 1491-1513. † Authors listed alphabetically.

[J4] George Stepaniants*, Alasdair D. Hastewell*, Dominic J. Skinner, Jan F. Totz, and Jörn Dunkel. "Discovering dynamics and parameters of nonlinear oscillatory and chaotic systems from partial observations." *Physical Review Research* 6, 043062 (2024). * Equal contribution.

[J5] Marie Breeur*, George Stepaniants*, Pekka Keski-Rahkonen, Philippe Rigollet, and Vivian Viallon. "Optimal transport for automatic alignment of untargeted metabolomic data." *eLife* 12:RP91597 (2024). * Equal contribution.

[J6] George Stepaniants. "Learning partial differential equations in reproducing kernel Hilbert spaces." *Journal of Machine Learning Research* 24.86 (2023): 1-72.

[J7] George Stepaniants, Bingni W. Brunton, and J. Nathan Kutz. "Inferring causal networks of dynamical systems through transient dynamics and perturbation." *Physical Review E* 102.4 (2020): 042309.

### Conference Proceedings

[C1] Enric Boix-Adserà, Hannah Lawrence, George Stepaniants, and Philippe Rigollet. "GULP: a prediction-based metric between representations." *Advances in Neural Information Processing Systems* (2022).

[C2] Sinho Chewi, Julien Clancy, Thibaut Le Gouic, Philippe Rigollet, † George Stepaniants, and Austin Stromme. "Fast and smooth interpolation on Wasserstein space." *International Conference on Artificial Intelligence and Statistics* PMLR (2021). † Authors listed alphabetically.

### In Preparation

[P1] David Darrow*, George Stepaniants*, and Chris Camaño. "SIEVE: Spectral integral transforms, poly-exponential approximants, and Volterra equations." * Equal contribution.

## Talks and Presentations

### Organized Symposia

- "Minisymposium on Data-Driven Methods for Multiscale Modeling and Homogenization", SIAM Conference on Computational Science and Engineering, Fort Worth, March 2025
- "Minisymposium on Data-Driven Learning of Dynamical Systems from Partial Observations", SIAM Conference on Mathematics of Data Science, Atlanta, October 2024

**Invited Talks**

- "A Shortcake Counterexample to the Planar Pompeiu and Schiffer Conjectures", Columbia Applied Mathematics Colloquium, New York, November 2026 **(Upcoming)**
- "Learning Memory and Material Dependent Constitutive Laws", Cambridge Engineering - Mechanics and Materials Seminar Series, Cambridge, October 2026
- "A Complete Resolution of Forsythe's Conjecture: An Exploration with AI-Assisted Mathematics", University of Manchester Centre for AI Fundamentals, Manchester, October 2026
- "Volterra Integral Equations and Memory Dependent Constitutive Laws", Imperial College London Dynamical Systems Seminar, London, October 2026
- "What We Did Last Summer: Adventures in AI-Assisted Mathematics", DAMTP Colloquium, Cambridge, October 2026
- "A Shortcake Counterexample to the Planar Pompeiu and Schiffer Conjectures", Imperial College London CCMI Seminar, London, October 2026
- "A Shortcake Counterexample to the Planar Pompeiu and Schiffer Conjectures", Princeton IDeAS Seminar, Princeton, September 2026
- "Volterra Integral Equations and Memory Dependent Constitutive Laws", Isaac Newton Institute, Cambridge, August 2026
- "Learning Material Constitutive Laws with Neural Operators", ILAS, Virginia Tech, May 2026
- "Learning Dynamics with Limited Data", UT Austin Oden Institute & Department of Statistics and Data Sciences, Online, January 2026
- "Volterra Integral Equations and Memory Dependent Constitutive Laws", UC Irvine Applied & Computational Math Seminar, Irvine, October 2025
- "Learning Memory and Material Dependent Constitutive Laws", Surrogates and Dimension Reduction in Scientific Machine Learning, Manchester University, September 2025
- "Alignment of Untargeted Data through their Covariances: A Novel Perspective on a Classical Tool in Optimal Transport", Joint Statistics Meeting, Nashville, August 2025 **(One of 3 PhD students selected for the prestigious IMS Lawrence D. Brown PhD Student Award)**
- "A Spectral Theory of Volterra Equations: Applications to Learning of Material Laws", Efficient and Reliable Deep Learning Methods and their Scientific Applications, **Banff BIRS Centre**, June 2025
- "Learning Dynamics of Hidden Variables in Multiscale Viscoelastic Materials", SIAM Conference on Applications of Dynamical Systems, Denver, May 2025
- "A Spectral Theory of Scalar Volterra Equations", Dartmouth Applied & Computational Math Seminar, Dartmouth, March 2025
- "A Spectral Theory of Scalar Volterra Equations", MIT Applied Math Physical Mathematics Seminar, Cambridge, March 2025
- "Learning Memory and Material Dependent Constitutive Laws", Differential Equations for Data Science, Kyoto University, February 2025
- "Discovering dynamics and parameters of nonlinear oscillatory and chaotic systems from partial observations", Fourth Symposium on Machine Learning and Dynamical Systems, **Fields Institute**, July 2024
- "Covariance Alignment with Optimal Transport", Yale Applied Mathematics Seminar, New Haven, April 2024
- "Gromov-Wasserstein Theory and Application to Metabolomics", SIAM Conference on Uncertainty Quantification, Trieste, Italy, February 2024
- "Gromov-Wasserstein Theory and Application to Metabolomics", Statistics and Learning Theory Summer School, Tsaghkadzor, Armenia, July 2023 **(One of 7 invited speakers)**
- "Optimal transport for automatic alignment of untargeted metabolomic data", Harvard Applied Math Graduate Student Seminar, Cambridge, March 2023
- "Learning PDEs in a Reproducing Kernel Hilbert Space", SIAM Conference on Mathematics of Data Science, San Diego, September 2022

- "Learning PDEs in a Reproducing Kernel Hilbert Space", Meeting on Mathematical Statistics, **CIRM, Marseille, France**, December 2021 **(Only 3 graduate student speakers invited)**

**Contributed Talks**

- "Discovering dynamics and parameters of nonlinear oscillatory and chaotic systems from partial observations", Dynamics Days, UC Davis, January 2024
- "Learning and predicting complex systems dynamics from single-variable observations", APS March Meeting, Chicago, March 2022
- "Learning PDEs in a Reproducing Kernel Hilbert Space", LIDS Stats & Tea, MIT, December 2021
- "Inferring causal networks of dynamical systems through transient dynamics and perturbation", Econometrics Lunch, MIT, December 2021
- "Fusion of Genetically Incompatible Fungal Cells", UCLA Computational and Applied Math REU Presentation, IPAM, August 2018
- "Quantifying Rupture Risk of Brain Aneurysms", MATDAT18: NSF Materials and Data Science Hackathon, Alexandria, June 2018
- "Hyperparameter Selection", AI2 Research Internship Final Presentation, Seattle, August 2017
- "Beaker Experimentation Platform", AI2 Research Internship Midterm Presentation, Seattle, August 2017
- "Image Analysis in Parkinson's Research", Pfizer Research Internship Final Presentation, Cambridge, August 2016

**Poster Presentations**

- "Covariance alignment: from maximum-likelihood estimation to Gromov-Wasserstein", Cornell ORIE Young Researchers Workshop, Cornell, October 2023
- "Inferring causal networks of dynamical systems through transient dynamics and perturbation", Undergraduate Research Symposium, UW, June 2019

## Teaching Experience

**(ACM 270) California Institute of Technology – Instructor of Record** — Spring 2025

Designer and Instructor of New Course for Data-Driven Modeling of Dynamical Systems

*Two weekly 1.5 hour classes with 30+ students, with final presentations and projects, unifying topics in system identification, sparse model inference, Koopman theory, adjoint methods, data assimilation, and scientific ML. Turning class content and lectures into a class text.*

**(MIT 18.032) Massachusetts Institute of Technology – Teaching Assistant** — Spring 2022

Teaching Assistant for Differential Equations (theory focused)

*Two weekly 1 hour recitations with 16 students, including grading of homeworks/finals and proctoring final exams. Covering existence and uniqueness theory (Cauchy-Lipschitz Theorem, Picard iterations), control and comparison of solutions, maximal solutions, Grönwall's Lemma, multivariable ODEs, and fixed point stability.*

**(MIT 18.600) Massachusetts Institute of Technology – Teaching Assistant** — Fall 2021

Teaching Assistant for Introduction to Probability

*Two weekly 1 hour recitations with 40 students, including grading of homeworks and finals. Covering probability spaces, random variables, classes of distributions, Bayes theorem, Chebyshev inequality, law of large numbers, and central limit theorems.*

## Mentoring Experience

**(SURF) California Institute of Technology** — Summer 2025

Undergraduate Research Mentor for Vladislav Syntko (Maastricht University, Netherlands)

A Signature-Based Approach for System Identification and Control: *Applications and theory for signature transform methods in open-loop control of dynamical systems.*

**(WAVE) California Institute of Technology** — Summer 2025

Undergraduate Research Mentor for Owen Tolbert (University of Maryland Baltimore County)

A Study of Network Inference Methods: *Information-theoretic and deep learning methods for inference of networked dynamical systems.*

**(MCM) California Institute of Technology** — Fall 2024

Trained three undergraduate Caltech teams for the Mathematical Contest in Modeling (MCM), coaching and solving practice problems over the course of several months

- (Honorable Mention) Gautham Kappaganthula, Constantin Cedillo-Vayson de Pradenne, Colin La
- (Successful Participant) Joseph Pieper, Sujay Champati, Dhruv Verma
- (Successful Participant) James Hou, Aman Burman, Abhiram Cherukupalli

**(Mentor) California Institute of Technology** — Fall 2024

Undergraduate Research Mentor for Zixiang Zhou (University of Southern California)

*Mentored research reading and project in data-driven dynamical systems inference algorithms based on the method of characteristics.*

**(SPUR+) Massachusetts Institute of Technology** — Summer 2023 – Fall 2023

Undergraduate Research Mentor for Elaine Liu (Massachusetts Institute of Technology)

Modeling International Trade and Tariffs: *Study of large trade and tariffs dataset across 200 world countries, investigating the use of spectral and graph wavelet decompositions for analysis of temporal trade network data.*

**(DRP) Massachusetts Institute of Technology** — Summer 2023

Directed Reading Program with Loreta Arzumanyan and Joshua Curtis Kuffour (Massachusetts Institute of Technology)

*Guided reading of two undergraduate students in the graduate dynamical systems text "Stability, Instability and Chaos" by Paul Glendinning over the course of the summer. Prepared students to present their knowledge of the text in a final presentation at the end of summer.*

**(Mentor) Massachusetts Institute of Technology** — Summer 2023

Undergraduate Research Mentor for Donald J. Liveoak and Hanna Chen (Massachusetts Institute of Technology)

*Guided research readings with two MIT undergraduates on optimal transport and adjoint methods for inference of stochastic dynamical systems and networked dynamical systems.*

**(UROP) Massachusetts Institute of Technology** — Fall 2021 – Spring 2022

Undergraduate Research Mentor for David Darrow (Massachusetts Institute of Technology)

Optimal Transport for Protein Folding: *Studying how optimal transport and Gromov-Wasserstein methods can be used to predict the three-dimensional structure of proteins.*

David Darrow awarded 2022 Churchill Scholarship

**(Community Service) University of Washington**

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| Math tutoring from K12 to college-level subjects | 2015 – 2019 |
| Teaching assistant at University of Washington Math Circle | 2015 – 2016 |

## Service and Leadership

1. Part III Subject Advisor for the ACDA subgroup, DAMTP, University of Cambridge 2026 – 2027
2. Organizer of the DAMTP ACDA Colloquium, University of Cambridge 2026 – 2027
3. Organizer of Caltech CMS Departmental CMX Seminar 2025 – 2026
4. Board Member of One World Seminar on Mathematics of Machine Learning 2025
5. Organizer of SURF/WAVE undergraduate research across three faculty at Caltech CMS 2025
6. Presenter on AI literacy and societal impact in LA and NY Armenian communities 2025
7. Organized panel on graduate school experience and research in Redmond Armenian community 2020
8. Founded and led Armenian Student Association at the University of Washington (ASAUW) 2015 – 2019
9. Competition judge at the University of Washington Math Olympiad 2015 – 2019

## Professional Membership

- Institute of Mathematical Statistics (IMS) 2024 – Present
- Society for Industrial and Applied Mathematics (SIAM) 2020 – Present
- American Physical Society (APS) 2019 – Present